2024

COMPUTER SCIENCE AND ENGINEERING

Paper : CSEL-711

(Internet of Things)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Group - A

Answer ***any three*** questions. 2×3

1. Explain, how IoT applications in connected vehicles contribute to road safety. (CO5)
2. 'IoT-enabled smart home devices enhance energy efficiency.'— Comment. (CO5)
3. Describe how IoT devices help monitor soil moisture and weather conditions to increase crop yield. (CO5)
4. Demonstrate how IoT-based sensors are used for monitoring air quality in smart cities. (CO5)

Group - B

Answer ***any two*** questions. 2×2

5. Illustrate Data in Motion vs. Data in Rest-Role. (CO4)
6. Explain importance of Hadoop in developing a large-scale IoT Infrastructure. (CO4)
7. Enlighten the role of Apache Spark in IoT-based solutions. (CO4)

Group - C

Answer ***any one*** question.

8. (a) Illustrate IoT Data Management and Compute Stack with Fog Computing.
   (b) Describe the characteristics of a smart object.
   (c) Point out the significant limitations of smart objects in WSN formation. (CO1) 5+3+4

9. (a) Explain importance of Distributed Computing and Data Management across an IoT System.
   (b) Discuss the functions of Edge and Cloud Computing layers in IoT reference model.
   (c) Illustrate the role of Gateway in IoT network. (CO1) 5+5+2

Group - D

Answer *any one* question.

10. (a) Illustrate LoRaWAN layers and their functions with a suitable diagram.

(b) Discuss the security issues dealt in LoRaWAN. (CO2) 8+4

11. (a) Illustrate IEEE 802.15.4g/e MAC Frame Format.

(b) Discuss the importance of Auxiliary Security Header in IEEE 802.15.4g/e. (CO2) 8+4

Group - E

Answer *any one* question.

12. (a) Compare an IoT Protocol Stack Utilizing 6LoWPAN over an IP Protocol Stack.

(b) Discuss IPv6 header compression technique for 6LoWPAN.

(c) Analyze the importance of fragmentation for IPv6 transmission over 6LoWPAN. (CO3) 5+4+3

13. (a) Explain SCADA Transport over LLNs with MAP-T.

(b) Compare CoAP and MQTT with respect to scalability issues.

(c) Why SCADA was adopted for IP? (CO3) 5+4+3

Group - F

Answer *any one* question.

14. (a) Discuss the need for Apache Kafka in real-time data processing.

(b) Illustrate the working principle of Map-Reduce frame work. (CO4) 6+6

15. (a) Discuss the importance of Apache Spark in processing data generated by IoT Infrastructure.

(b) Illustrate Lambda Architecture where Apache Spark, Apache Kafka and Hadoop work together to generate a consolidated view. (CO4) 6+6

Group - G

Answer *any one* question.

16. (a) Design a smart connected parking solution for a Smart City that can reduce road traffic and average fuel consumption.

(b) Analyze the need for cloud services in integration of parking services provided by different vendors. (CO5) 8+4

17. (a) Design Street Lighting Architecture for a Smart City that can be controlled by operators from a remote location. Moreover, the system should automatically be enabled, if the sky is very cloudy.

(b) "IoT can help in traffic management in a Smart City."— Justify the statement. (CO5) 8+4